Barinder Singh Banwait

Personal Action Plan · BlueDot Impact AGI Strategy course · October 2026

Summary

I have decided to work on agreement verification technology, and more broadly, trying to build "provably safe AI", as defined and discussed by Max Tegmark here.

Note that "provable" here is not in the legal sense, but rather the mathematical sense. Here, it means an AI system that has passed a formal verification protocol. This is connected to this Tailwind project call. Formal verification is the "gold standard" of correctness.

I am uniquely positioned to contribute because of my strong mathematical research background, which in recent years has in fact involved significant Lean 4 development (Lean 4 is a formal verification language now widely used in math). I have also worked in surgical robotics in a previous role, on low-level C code that had to pass lots of regulatory requirements (IEC 62304) pertaining to software on medical devices. So I know something about ensuring a hardware system meets its requirements. And this is relevant to the Tailwind description. Also my mathematical background on elliptic curves is strongly connected to elliptic curve cryptography, and I have worked in a cryptography research lab before (INRIA Bordeaux). Since the description says "will need to include hardware engineers, cryptographers and formal verification researchers", I think it speaks to a lot of my skills and experience.

As said below, I'll be doing the Iliad Fellowship from November.

Strategic assessment

I want to work towards agreement verification technology and building provably safe AI.

The deployment of the technology this project enables will require certain other world assumptions to be met: mainly that frontier labs in the US and China all agree on a collective slowdown. CEOs of such companies and loads of the engineers working there have already called for this. I'm hoping, and assuming, that this will happen.

Once it does, then as per Plan A in AI 2040, different parties will need to verify that the other parties aren't violating the agreements on e.g. inference-only compute, not new training runs. This is one small part of the much wider agreement on collective slowdown.

If nobody acts, then it's unclear why a slowdown will ever happen. That means exponential development and recursive self-improvement of multiple AI systems, some of which will most likely not be aligned to human values in some way; this then opens the way to the multiple doom scenarios we learned about in the course.

Why this intervention? Because basically nobody is working on it. And it's also a fascinating multi-disciplinary challenge bringing together loads of subjects towards something of great value to the world. I've also personally worked on many of these things in my past experiences; I explain this in the summary above.

Theory of change

Slightly different to the above scenario, this is about trying to ensure that an individual agent is safe:

If I make a "Lean kernel" for an AI agent, then it can be used [this is the most risky assumption] to check the agent is verified safe, which leads to the agent being guaranteed to refuse to tell the user how to engineer a dirty bioweapon, assuming:

- we can create the environment needed to "check the AI against the kernel"
- we can create the kernel itself

The riskiest assumption will require lots of red-teaming checks, different groups trying to get the system to tell the user how to do bad, evil things like building a backyard nuclear weapon or <enter bad thing here>.

Concrete commitments

I'll be joining the Iliad Fellowship from November 2026–February 2027; this is a technical AI safety fellowship towards exactly what I want to be doing.